

EXPERIMENT-14:

Implement House price Prediction using appropriate machine learning algorithm.

Code:

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# =====  
# House Price Prediction using Linear Regression  
# Dataset: data.csv  
# =====  
  
# Step 1: Upload Dataset  
from google.colab import files  
uploaded = files.upload()  
  
# Step 2: Import Libraries  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression  
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
  
# Step 3: Load Dataset  
df = pd.read_csv("data.csv")  
print("First 5 Rows:")  
print(df.head())  
print("\nDataset Shape:", df.shape)  
  
# Step 4: Display Column Names  
print("\nColumns:")  
print(df.columns)  
  
# Step 5: Remove unnecessary columns (if present)  
drop_cols = ["date", "street", "country", "statezip"]  
for col in drop_cols:  
    if col in df.columns:  
        df.drop(col, axis=1, inplace=True)
```

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# Step 6: Convert Categorical Columns to Numeric
df = pd.get_dummies(df, drop_first=True)

# Step 7: Handle Missing Values
df.fillna(df.median(numeric_only=True), inplace=True)

# Step 8: Features and Target
X = df.drop("price", axis=1)
y = df["price"]

# Step 9: Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

# Step 10: Train Model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 11: Predict
y_pred = model.predict(X_test)

# Step 12: Evaluation
print("\nModel Performance")
print("-----")
print("R2 Score :", round(r2_score(y_test, y_pred),4))
print("MAE      :", round(mean_absolute_error(y_test, y_pred),2))
print("MSE      :", round(mean_squared_error(y_test, y_pred),2))
print("RMSE     :", round(np.sqrt(mean_squared_error(y_test, y_pred)),2))

# Step 13: Actual vs Predicted Graph
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")

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plt.ylabel("Predicted Price")
plt.title("House Price Prediction")
plt.grid(True)
plt.show()

# Step 14: Predict First House
sample = X.iloc[[0]]
prediction = model.predict(sample)
print("\nPredicted House Price:")
print("₹", round(prediction[0],2))

```

Output:

```

... Choose Files data.csv
data.csv(text/csv) - 526795 bytes, last modified: 8/5/2026 - 100% done
Saving data.csv to data.csv
First 5 Rows:

```

	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	\
0	2014-05-02 00:00:00	313000.0	3.0	1.50	1340	7912	
1	2014-05-02 00:00:00	2384000.0	5.0	2.50	3650	9050	
2	2014-05-02 00:00:00	342000.0	3.0	2.00	1930	11947	
3	2014-05-02 00:00:00	420000.0	3.0	2.25	2000	8030	
4	2014-05-02 00:00:00	550000.0	4.0	2.50	1940	10500	

	floors	waterfront	view	condition	sqft_above	sqft_basement	yr_built	\
0	1.5	0	0	3	1340	0	1955	
1	2.0	0	4	5	3370	280	1921	
2	1.0	0	0	4	1930	0	1966	
3	1.0	0	0	4	1000	1000	1963	
4	1.0	0	0	4	1140	800	1976	

	yr_renovated	street	city	statezip	country
0	2005	18810 Densmore Ave N	Shoreline	WA 98133	USA
1	0	709 W Blaine St	Seattle	WA 98119	USA
2	0	26206-26214 143rd Ave SE	Kent	WA 98042	USA
3	0	857 170th Pl NE	Bellevue	WA 98008	USA
4	1992	9105 170th Ave NE	Redmond	WA 98052	USA

Dataset Shape: (4600, 18)

Columns:

```

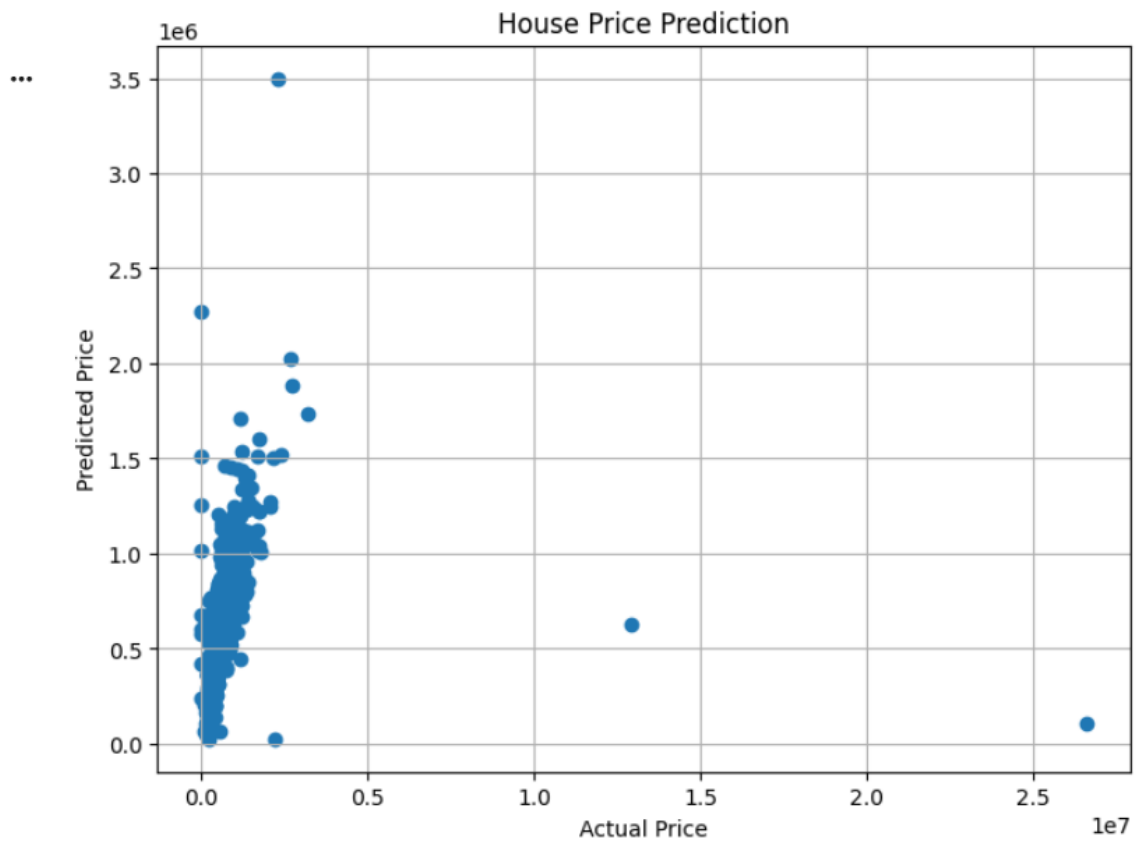
Index(['date', 'price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot',
      'floors', 'waterfront', 'view', 'condition', 'sqft_above',
      'sqft_basement', 'yr_built', 'yr_renovated', 'street', 'city',

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    'sqft_basement', 'yr_built', 'yr_renovated', 'street', 'city',  
    'statezip', 'country'],  
    dtype='object')
```

Model Performance

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R2 Score : 0.0415  
MAE       : 172954.68  
MSE       : 977554817805.59  
RMSE      : 988713.72
```



Predicted House Price:
₹ 312716.06